

Anthony Cabrera

HETEROGENEOUS COMPUTING RESEARCH SCIENTIST

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Summary

Experienced classical and quantum computing and compiler researcher with a penchant for exploring questions around the hardware-software interface. Open-source contributor to the LLVM project and LLVM Test Suite. Adept at quickly learning skills, languages, or tools necessary to answer research questions. Strives to communicate complex ideas effectively and accessibly.

Research and Work Experience

Quantum Circuits, Inc.

New Haven, CT (Remote)

QUANTUM SOFTWARE ENGINEER - COMPILERS

December 2024 - Present

- Leading the design and implementation of the compiler back-end for Quantum Circuits QPUs
- Integrating the compiler with custom, RF-range qubit hardware controllers
- Implemented emulator of real-time processor controller for verification of back-end compiler code generation

Architectures and Performance Group @ Oak Ridge National Laboratory

Oak Ridge, TN (Remote)

RESEARCH SCIENTIST

Jan 2022 - Present

SOFTWARE ENGINEER

Aug 2020 - December 2021

- Created an MLIR pass to enable qubit-wise commutativity optimizations in the Xanadu Catalyst MLIR quantum compiler
- Deployed TF/PyTorch models on AMD/Xilinx Versal ACAP VCK190 platform for scientific application edge computing
- Upstreamed MLIR dialect and lowering features for Flang – the LLVM front-end for Fortran
- Led a multi-institution performance and portability evaluation comparing Intel and Xilinx FPGA OpenCL kernels
- Developed Hexagon DSP kernels for the Qualcomm Snapdragon chip, as part of DARPA's DSSoC project
- Created performance models for CXL-based GPU-FPGA collaboration on HPC mini applications as part of the DoE Exascale Compute Project

Computer Science and Engineering Department @ Washington University in St. Louis

St. Louis, MO

ADJUNCT INSTRUCTOR

August 2024 - Present

VISITING RESEARCH SCIENTIST

May 2022 - Present

- Teaching a graduate level course called Accelerating Algorithms in Reconfigurable Logic
- Led seminar presentations and contributed to research proposals in the areas of hardware and compilers

Stream Based Supercomputing Laboratory @ Washington University in St. Louis

St. Louis, MO

GRADUATE RESEARCH ASSISTANT

July 2016 - July 2020

- Thesis: Domain Specific Computing in Tightly Coupled Heterogeneous Systems
- Evaluated the Intel HARPv2 CPU+FPGA platform as a domain specific compute solution
- Created a benchmark suite of data integration applications (DIBS) to identify opportunities for hardware acceleration
- Ported Needleman-Wunsch OpenCL Kernels to the Intel HARPv2 CPU+FPGA platform to analyze kernel design, performance, and portability
- Architected and optimized hardware for DIBS applications using OpenCL targeting the Intel HARPv2

The MITRE Corporation

Shiloh, IL

GRADUATE PROTOTYPING AND SOFTWARE ENGINEER

May 2019 - August 2019

- Created a neural network to detect cars from wireless iPhone camera stream targeting the NVIDIA Jetson Nano
- Deployed containers on GPU-enabled HPC resources to train convolutional neural networks
- Mentored undergraduate intern project on hyperparameter performance analysis
- Maintained GitLab repository to document work and enable continued development of project
- Selected as one of four interns across all of MITRE's sites to deliver company-wide presentation on project

Arm Holdings

Austin, TX

GRADUATE RESEARCH INTERN

May. 2018 - Aug. 2018

- Quantified spatial and temporal locality by creating a novel technique based on reuse distance
- Developed dynamic binary instrumentation clients to profile memory subsystem characteristics
- Identified strategies around data layout transformations and paging to improve memory subsystem performance

Advanced Sensors Research Laboratory @ Washington University in St. Louis

UNDERGRADUATE AND GRADUATE RESEARCH ASSISTANT

St. Louis, MO

May 2014 - July 2016

- Developed the software/UI for filter alignment of NIR fluorescence imagers
- Assisted medical researchers with NIR fluorescence and polarization imaging studies
- Aided in the design and fabrication of a custom PCB around an ultra low-noise imaging sensor

Education

Washington University in St. Louis

PHD COMPUTER ENGINEERING

MS COMPUTER SCIENCE

BS COMPUTER ENGINEERING, SECOND MAJOR COMPUTER SCIENCE

BSAS ELECTRICAL ENGINEERING

St. Louis, MO

August 2020

August 2018

May 2015

Hendrix College

BA CHEMICAL PHYSICS, MINOR MUSIC

Conway, AR

May 2013

Peer Reviewed Publications (Chronological)

1. AJ Adams, S Khan, AS Bhamra, RR Abusaada, **AM Cabrera**, CC Hoechst, TS Humble, JS Young, TM Conte "ASDF: A Compiler for Qwerty, a Basis-Oriented Quantum Programming Language", *IEEE/ACM International Symposium on Code Generation and Optimization (CGO '25)*
2. **AM Cabrera**, S Afrose, D Claudino, TS Humble "Toward Exploiting Qubit-Wise Commutativity Using an MLIR Approach", *In Preparation*.
3. **AM Cabrera**, DE Bernholdt, C Zimmer "A Flang Plugin for Fortran Feature Characterization", Accepted for publication in *IEEE/ACM Eighth Workshop on the LLVM Compiler Infrastructure in HPC (LLVM-HPC '24)*.
4. E Wong, VL Ortega, D Claudino, S Johnson, S Afrose, M Gowrishankar, **AM Cabrera**, TS Humble "A Cross-Platform Execution Engine for the Quantum Intermediate Representation", *arXiv preprint arXiv:2404.14299*.
5. **AM Cabrera**, YA Yucesan, FY Liu, W Blokland, JS Vetter "Errant Beam Detection Using the AMD Versal ACAP and Vitis AI", *IEEE High Performance Extreme Computing Conference (HPEC '23)*.
6. CJ Faber, SD Harris, Z Xiao, RD Chamberlain, **AM Cabrera** "Challenges Designing for FPGAs Using High-Level Synthesis", *IEEE High Performance Extreme Computing Conference (HPEC '22)*.
7. NR Miniskar, AR Young, FY Liu, **AM Cabrera**, JS Vetter, "Ultra Low Latency Machine Learning for Scientific Edge Applications", *IEEE International Conference on Field Programmable Logic and Applications (FPL '22)*.
8. AR Young*, **AM Cabrera***, JS Vetter, "Design and Analysis of CXL Performance Models for Tightly-Coupled Heterogeneous Computing", *ACM International Workshop on Extreme Heterogeneity Solutions (ExHET '22 @ PPoPP '22)*.
9. CJ Faber, T Plano, S Kodali, Z Xiao, A Dwaraki, JD Buhler, RD Chamberlain, **AM Cabrera**, "Platform Agnostic Streaming Data Application Performance Models", *ACM/IEEE Redefining Scalability for Diversely Heterogeneous Architectures (RSDHA '21 @ SC '21)*.
10. Zhili Xiao, RD Chamberlain, **AM Cabrera**. "HLS Portability from Intel to Xilinx: A Case Study", *IEEE High Performance Extreme Computing Conference (HPEC '21)*. [\[Paper\]](#) [\[Slides\]](#)
11. **AM Cabrera**, S Hitefield, J Kim, S Lee, NR Miniskar, JS Vetter, "Toward Performance Portable Programming for Heterogeneous System-on-Chips: Case Study with Qualcomm Snapdragon SoC", *IEEE High Performance Extreme Computing Conference (HPEC '21)*. [\[Paper\]](#) [\[Slides\]](#)
12. **AM Cabrera**, AR Young, J Lambert, Z Xiao, A An, S Lee, Z Jin, J Kim, J Buhler, RD Chamberlain, JS Vetter, "Toward Evaluating High-Level Synthesis Portability and Performance between Intel and Xilinx FPGAs", *ACM International Workshop on OpenCL (IWOCL '21)*. [\[Paper\]](#) [\[Slides\]](#) [\[Video\]](#)
13. **AM Cabrera**, RD Chamberlain, "Design and Performance Evaluation of Optimizations for OpenCL FPGA Kernels", *IEEE High Performance Extreme Computing Conference (HPEC '20)*. [\[Paper\]](#) [\[Slides\]](#)
14. **AM Cabrera**, RD Chamberlain, "Designing Domain Specific Computing Systems", *IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM '20)*. [\[Paper\]](#) [\[Slides\]](#) [\[Video\]](#)
15. **AM Cabrera**, RD Chamberlain, JC Beard, "Multi-spectral Reuse Distance: Divining Spatial Information from Temporal Data", *IEEE High Performance Extreme Computing Conference (HPEC '19)*. [\[Paper\]](#) [\[Slides\]](#)
16. **AM Cabrera**, RD Chamberlain, "Exploring Portability and Performance of OpenCL FPGA Kernels on Intel HARPv2", *ACM International Workshop on OpenCL (IWOCL '19)* **Best Presentation Award**. [\[Paper\]](#) [\[Slides\]](#)
17. CJ Faber, **AM Cabrera**, O Booker, G Maayan, RD Chamberlain, "Data Integration Tasks on Heterogeneous Systems Using OpenCL", *ACM International Workshop on OpenCL (IWOCL '19)*. [\[Paper\]](#)

18. **AM Cabrera**, CJ Faber, K Cepeda, R Derber, C Epstein, J Zheng, RK Cytron, RD Chamberlain, "DIBS: A Data Integration Benchmark Suite", *ACM/SPEC International Conference on Performance Engineering (ICPE '18)*. [\[Paper\]](#) [\[Slides\]](#)

*Denotes equal contribution.

Presentations

2024 **Integrating an MLIR Pass into the Catalyst MLIR Compiler to Close the Loop for Qubit-Wise Commutativity**, QCE Quantum Software 2.0 Workshop [\[Slides\]](#) *Montreal, QC, CA*

PhD Dissertation

Domain Specific Computing in Tightly-Coupled Heterogeneous Systems [\[Text\]](#) [\[Slides\]](#)

Honors and Awards

2020	SC20 Early Career Program , Supercomputing 2020	<i>Atlanta, GA</i>
2020	Honors Designation for PhD Progress Review (Top 15-20% of students) , CSE Department @ WUSTL	<i>St. Louis, MO</i>
2020	Engineering PhD Student Commencement Marshal , WUSTL	<i>St. Louis, MO</i>
2019	Best Presentation Award , International Workshop on OpenCL	<i>Boston, MA</i>
2019	Graduate Student Ambassador , Intel Corporation	<i>St. Louis, MO</i>
2019	Travel Grant , Supercomputing 2019	<i>Denver, CO</i>
2018	Travel Grant , Supercomputing 2018	<i>Dallas, TX</i>
2017	Travel Grant , Supercomputing 2017	<i>Denver, CO</i>
2015	Graduate Danforth Scholar , WUSTL	<i>St. Louis, MO</i>
2013	Harold P. Brown Engineering Fellowship , McKelvey School of Engineering @ WUSTL	<i>St. Louis, MO</i>
2013	Hendrix College Chamber Orchestra Award , Hendrix College	<i>Conway, AR</i>
2012	Transamerica Employer Solution & Pension Scholarship Award , Transamerica Corporation	<i>Little Rock, AR</i>
2011	Hendrix College Chamber Orchestra Award , Hendrix College	<i>Conway, AR</i>

Teaching Experience

FL24	WUSTL CSE 565M Accelerating Algorithms in Reconfigurable Logic , Instructor of Record	<i>St. Louis, MO</i>
FL17, FL18	WUSTL CSE 560M Computer Systems Architecture I , Graduate Teaching Assistant	<i>St. Louis, MO</i>
SU18	WUSTL CSE 566S High Performance Computing , Graduate Teaching Assistant	
SP16	WUSTL CSE {4,5}63M Digital Integrated Circuit Design and Architecture , Graduate Teaching Assistant	
FL14, SP15	WUSTL CSE 200 Scientific Computing , Undergraduate Teaching Assistant	

Press

2021	SCTV Interview , SC21 Inclusion and Diversity with AJ Lauer and Anthony Cabrera	<i>St. Louis</i>
2021	SCTV Interview , Promoting the SCALE students program at SC	<i>Virtual</i>

Professional Service

2022	SC22 Inclusivity Committee , Supercomputing 2022	<i>Dallas, TX</i>
2022	Program Committee , International Workshop on OpenCL and SYCLcon	<i>Remote</i>
2021	SC21 Inclusivity Committee , Supercomputing 2021	<i>St. Louis, MO</i>
2021	Program Committee , International Workshop on OpenCL and SYCLcon	<i>Remote</i>
2019	Lead Student Volunteer: Communications Committee Press Liaison , Supercomputing 2019	<i>Denver, CO</i>
2019	Student Volunteer , Supercomputing 2018	<i>Dallas, TX</i>
2017	Student Volunteer , Supercomputing 2017	<i>Denver, CO</i>

Skills

- Languages** Bash, C, C++, Python
- Frameworks** CMake, Git, LLVM, MLIR, OpenCL, Xilinx HLS